B.E. COMPUTER SCIENCE ENGINEERING

PROJECT REVIEW 1

VERA — Verified Emergency & Route Advisory

AI-assisted, context-aware and crowd-verified route safety advisory platform.

Team
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Introduction

- Conventional navigation applications mainly focus on distance, travel time and traffic.
- The fastest route may not always be the safest route.
- Route safety can vary according to infrastructure, location, time and local conditions.
- Important safety factors include lighting, pedestrian activity, road isolation, incident history, transport accessibility and emergency accessibility.
- VERA proposes a route safety advisory layer that considers safety along with travel efficiency.

The Problem

Women commuting in Indian cities lack a practical, real-time way to know whether a route or stop is safe at a specific time. Navigation apps optimize for speed or distance; safety apps are reactive (SOS after an incident). Commuters need a proactive answer to: “Which route is actually safer right now, and why?”





Why This Matters

Millions of women depend on buses, metros, shared autos, and walking — often during early mornings or late evenings. Useful safety signals exist (streetlights, police posts, transit frequency) but are fragmented. Without a unified, time-sensitive view, commuters make decisions blind to real safety conditions.

- High exposure during off-peak hours
- Infrastructure data is under-utilized
- Current tools lack proactive, contextual safety guidance



Our Solution

VERA adds a dynamic safety layer on top of standard maps. Routes are color-coded (green/yellow/red) using a real-time safety score that blends infrastructure data, time-of-day adjustments, and verified crowd reports. The app helps commuters choose safer routes before they travel — proactive guidance, not just emergencies.

Objectives & approach

Objectives

- Build a safety-scoring engine for routes using streetlight density, police proximity, and transit frequency
- Design a time-of-day adjustment model — the same route scores differently at 8 AM vs 10 PM
- Build a verified, reputation-weighted crowd-reporting system with outlier detection
- Deliver a working mobile app with colour-coded routing and a one-tap SOS
- Validate the model against real Hyderabad corridors

Approach

VERA adds a route-safety scoring layer on top of standard maps — combining public infrastructure data, time-of-day risk, and verified crowd-sourced input, wrapped in a moderation pipeline that keeps the data trustworthy.

LITERATURE SURVEY

Research supports safety–efficiency trade-offs

Paper	Main focus	Relevance to VERA
Sohrabi et al. (2022), Safe route-finding: A review of literature and future directions	Safe route-finding must consider safety alongside conventional route efficiency; safety and travel time can trade off.	Supports multi-factor scoring and fastest-vs-safer comparison.
Purchase & Breetzke (2026), Danger by design? Mapping safe route choices in the city of Tshwane, South Africa	Safe and shortest routes can differ; historical incidents, safety infrastructure and time/context affect route safety.	Supports context-aware route-level assessment.

EXISTING SYSTEM

Different systems optimize different priorities

Current systems address movement or safety signals separately; an integrated route-level safe framework remains missing.

MOVEMENT

Conventional Navigation

Primarily focuses on shortest or fastest routes using distance, traffic and travel time. Safety is not the primary route-selection criterion.

SAFETY SIGNALS

Existing Safety-Oriented Platforms

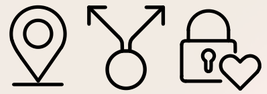
Provide safety scores, community information or emergency-support features, demonstrating the value of safety-related location information.

GAP IDENTIFIED

An integrated route-level framework is needed to combine multiple safety factors, contextual conditions, reliability-aware community observations and explainable route recommendations.

PROPOSED SYSTEM

From route request to safer action



Request & inputs

Route request — User location destination.

Alternatives — Generate alternative routes.

Safety factors — Collect and check infrastructure, incidents, activity, isolation, transport, emergency access, time/context and community reports.



Validate & compare

Reliability checkpoint — Check reliability and outliers.

Scoring engine — Calculate the safety score.

Compare — Compare routes on safety and efficiency.



Recommend & act

Explain — Explain the recommendation.

Safer route — Show the safer route and nearby emergency support.

Emergency action — Enable SOS and location sharing.

What makes VERA different

Novel contributions

- Dynamic safety scoring — recalculated by time of day, not a static label
- Blended trust model — objective data + reputation-weighted crowd input
- Outlier & fraud detection — spam bursts quarantined before reaching live scores
- Privacy-first — no continuous tracking, location captured only at report/SOS moments

Technology stack

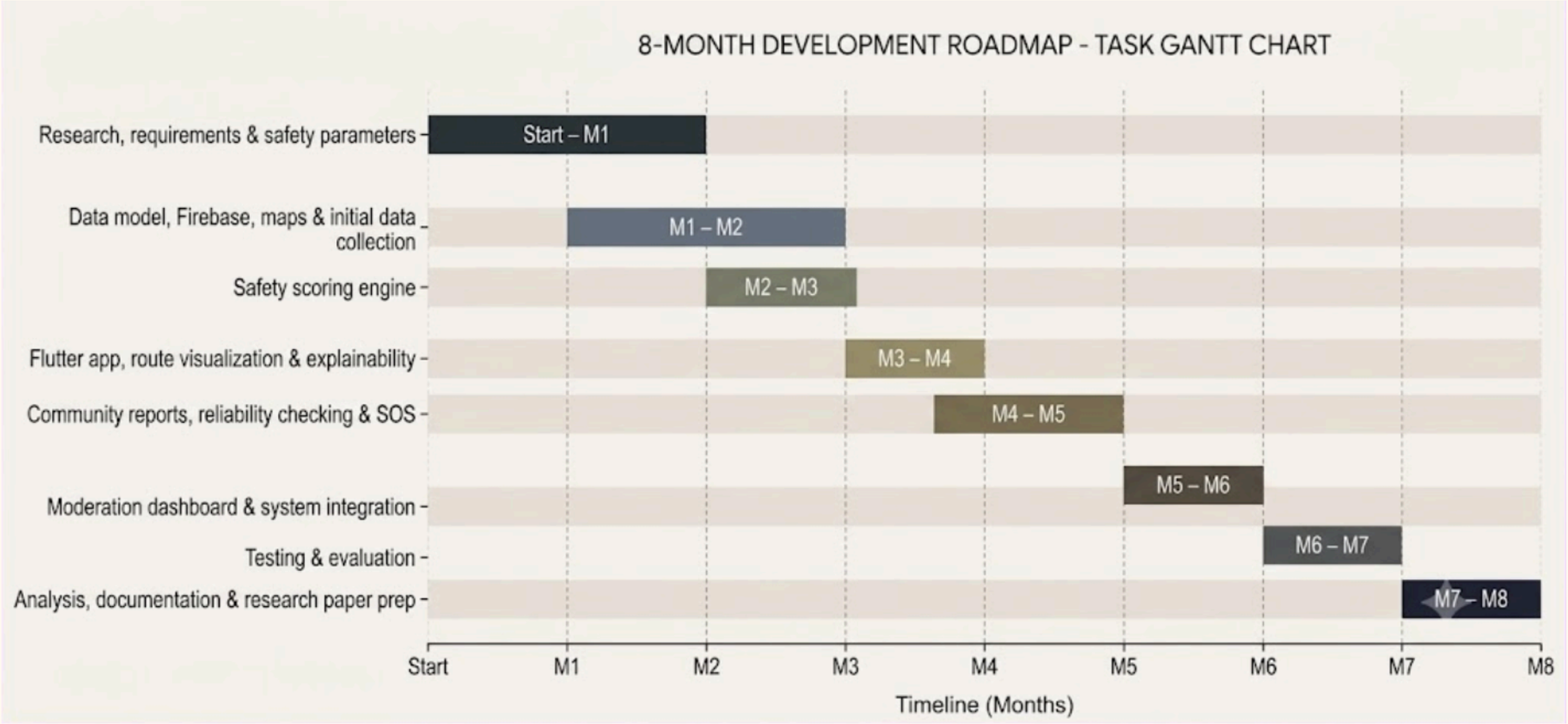
Layer	Tool
Mobile App	Flutter
Backend / DB	Firebase (Firestore, Auth, Functions)
Maps & Routing	Google Maps / OpenStreetMap
Scoring Engine	Node.js (weighted scoring model)
Moderation Dashboard	React

Methodology



The workflow combines multiple route safety factors, contextual conditions and reliability-checked community observations to calculate an explainable safety score and recommend a safer route.

PROJECT TIMELINE



Current progress

30

automated tests passing

- Scoring engine — time-of-day adjustment, reputation-weighted blending, spam quarantine
- Routing engine — safest-vs-fastest pathfinding, tested end to end
- Data pipeline — OSM, streetlight, and GTFS transit joins, tested
- Moderation dashboard — builds clean, ready for live data
- Flutter app — running on an Android emulator, connected to the Firebase backend

In progress: importing real Tolichowki corridor data (street geometry, lighting, transit) into the live backend

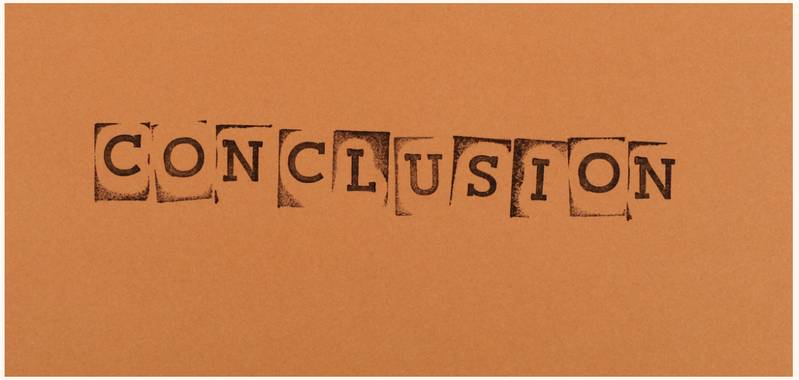
CURRENT PROJECT SCOPE

- Multi-factor route assessment
- Context-aware safety scoring
- Community-based reporting
- Incident/Reliability checking
- Actionable safety insights
- Nearby emergency support
- Critical SOS features
- Location and scope evaluation

API

FUTURE PROJECT SCOPE

- Predictive AI-based safety assessment
- Real-time info weather/ weather, event, and crowd information
- Advanced ML-based report fraud detection
- Route deviation/ unexpected-stop detection
- Driver/transport verification
- Voice emergency assistance
- Walk-with-me safety support
- API integrations



- VERA extends conventional navigation by considering route safety together with travel efficiency.
- It combines multiple safety factors into a route-level safety assessment.
- Time/context and community observations improve the contextual nature of the assessment.
- Reliability checking helps reduce the effect of potentially misleading reports.
- Explainable scoring helps users understand route recommendations. Fastest-vs-safer comparison makes the safety/travel-time trade-off clear.
- The project will be evaluated using selected Hyderabad routes. Final statement: “VERA aims to make route selection more safety-aware, explainable and context-sensitive.”

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